Herpes Zoster (HZ): Pathogenesis, Epidemiology, and Guideline Overview

When virus-specific cellular immunity declines during ageing or due to immunosuppression, latent varicella-zoster virus (VZV) can reactivate, leading to herpes zoster (HZ). Reactivation typically occurs in one or more dorsal root ganglia (DRG). Following reactivation, virions are transported antidromically along axons via the microtubular system. Upon reaching intra-epidermal nerve endings and the perifollicular neural network, viral replication is initiated in epidermal and/or infundibular keratinocytes. This replication disrupts normal keratinocyte differentiation, inducing gene expression patterns associated with blistering and vesicle formation. Histologically, cytopathic effects are evident, including multinucleated giant cells, syncytia, eosinophilic nuclear inclusions, and eventual apoptosis of infected cells.

Epidemiology and Clinical Impact

The incidence of HZ is 2–3 per 1,000 person-years in the general population, increasing to 7–10 per 1,000 person-years after age 50. In Spain, the hospitalization rate due to HZ is approximately 10 per 100,000 per year, and the disease significantly impairs patients' quality of life. Incidence rises markedly with age and is further elevated in individuals with immunodeficiency.

A common and often treatment-resistant complication of HZ is postherpetic neuralgia (PHN). While overall mortality from HZ is low in European countries, it can reach up to 19.5 per 100,000 in individuals over 95 years of age.

The introduction of a vaccine for HZ prevention has demonstrated a 51% reduction in HZ incidence. However, there is insufficient evidence to confirm a specific reduction in PHN incidence beyond what is achieved through lowering

HZ occurrence. Given that the lifetime risk of HZ in unvaccinated individuals by age 85 is estimated at 50%, the burden of HZ remains substantial even in vaccinated populations.

Scope, Purpose, and Methodology

This clinical guideline was developed in accordance with the quality criteria outlined in the AGREE II Instrument. Full details on scope, purpose, and methodology are provided in the accompanying methods report (available as an online supplement).

Recommendations are categorized into five levels of strength, indicated by both wording and symbols: - Strong recommendation in favor: ↑↑ - Weak recommendation in favor: ↑ - No recommendation: 0 - Weak recommendation against: ↓ - Strong recommendation against: ↓↓

Each recommendation includes the level of agreement among the expert panel (≥50%, ≥75%, ≥90%).

Guidelines require regular updating to reflect evolving evidence. This version is valid until June 2021. It may become outdated earlier if significant changes occur in evidence, clinical practice, available treatments, or drug approvals.

The first part of this guideline focuses on diagnostic approaches in patients presenting with suspected HZ. The section was authored by A. F. Nikkels (lead), J. Breuer, G. E. Gross, R. Lapid-Gortzak, U. Pleyer, G. M. Verjans, P. Wutzler, A. M. Agius, T. M. Lesser, and J. Sellner. Final recommendations were formally approved by the guideline's expert panel.

General Clinical Considerations

Herpes zoster classically presents as a unilateral, dermatomal rash. Skin lesions progress synchronously through stages: from erythematous macules to papules, vesicles, pustules, and finally crusting, which typically occurs within 5–7 days. Not all segments of the affected dermatome are necessarily involved.